

InterviewAce AI – Project Report

Problem Statement

Preparing for job interviews is often challenging for students and job seekers due to limited access to personalized guidance, role-specific interview practice, and constructive feedback. Traditional interview preparation methods rely on static question banks or expensive mock interviews that are not adaptive. This project aims to develop an AI-powered Interview Trainer Agent using IBM watsonx.ai and Granite models that generates personalized interview questions, evaluates user responses, provides scores, identifies strengths and weaknesses, and recommends improvements to enhance interview readiness.

Proposed Solution

InterviewAce AI is built using Python, Streamlit, IBM watsonx.ai, and IBM Granite Models. Users select a job role and experience level, receive AI-generated interview questions, submit answers, and receive instant evaluation. The system provides performance scores, detailed feedback, model answers, and personalized learning recommendations. Session history enables continuous interview practice while an interactive dashboard tracks user progress.

Technology Used

- IBM watsonx.ai
- IBM Granite Model (Granite-3.2/Granite-4)
- IBM Cloud Lite
- Python
- Streamlit
- GitHub
- Streamlit Session State

Key Features

- Personalized interview question generation
- AI-based answer evaluation and scoring
- Role-based interview preparation
- Improvement suggestions and model answers
- Progress tracking dashboard
- Scalable deployment-ready architecture

Benefits

- Automates interview preparation with AI assistance.
- Provides consistent and unbiased evaluation.
- Identifies strengths and skill gaps.
- Supports multiple job roles and experience levels.
- Improves confidence and interview readiness.